

Lecture 23 Scribe



L23_Amazon Sale_2026_...
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LECTURE 23 SCRIBE

Predicting the Impossible: Tail Bounds and System Reliability

(Case Study: Amazon Black Friday Sale 2026)

◆ Slide 1: Introduction

Topic

Predicting the Impossible: Tail Bounds and System Reliability

Context

- Case Study: **Amazon Black Friday Sale 2026**
- Focus: Understanding uncertainty in large-scale systems using probabilistic bounds

Instructor

- Dhaval Patel, PhD

- Associate Professor, CSE
 - MICxN Research Lab
 - SEAS, Ahmedabad University
-

◆ Slide 2: Amazon Black Friday Sale 2026

Overview

- Expected Start Date: **Friday, November 27, 2026**
- Based on:
 - Historical trends
 - Forecasts from industry sources

Key Idea

- Large-scale online sales involve **massive uncertainty**
 - Systems must handle:
 - Sudden spikes in user traffic
 - High transaction loads
 - Risk of system failure
-

◆ Slide 3: Computing Under Uncertainty

Concept

Amazon sale scenario is used to understand:

- **Uncertainty in demand**
- **Random behavior of users**
- **System reliability under stress**

Core Question

How can we predict extreme events (like overload or failure) when exact prediction is impossible?

◆ Slide 4: Motivation

Problem Statement

- Exact prediction of system behavior is difficult

- Need tools to:
 - Estimate worst-case scenarios
 - Bound probabilities of extreme events

Solution Approach

Use **Tail Bounds** from probability theory.

◆ Slide 5: Tail Bounds

Definition

Tail bounds provide **upper limits** on the probability that a random variable deviates significantly from its expected value.

Purpose

- Measure **risk of extreme outcomes**
 - Ensure **system reliability**
-

◆ Slide 6: Key Tail Bounds Covered

1. Markov's Inequality
2. Chebyshev's Inequality
3. Chernoff Bound

These are used to estimate probabilities of rare or extreme events.

◆ Slide 7: Markov's Inequality

Definition

For a non-negative random variable X and $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Assumptions

- $X \geq 0$
- Expectation $E[X]$ exists

Interpretation

- Gives a **simple upper bound**
 - Does not require variance
-

◆ Slide 8: Significance of Markov's Inequality

- Works with minimal information
 - Useful when only **mean** is known
 - Provides **loose bound**
-

◆ Slide 9: Chebyshev's Inequality

Definition

For any random variable X with mean μ and variance σ^2 :

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Assumptions

- Mean and variance exist
-

◆ Slide 10: Interpretation of Chebyshev's Inequality

- Measures deviation from mean
 - Applies to **any distribution**
 - Gives tighter bound than Markov
-

◆ Slide 11: Importance of Chebyshev's Inequality

- Helps in:
 - Variance-based analysis
 - Risk estimation
 - Widely used in reliability analysis
-

◆ Slide 12: Chernoff Bound

Concept

Provides **exponentially decreasing bounds** for sums of independent random variables.

Use Case

- When dealing with:
 - Large datasets
 - Independent trials
-

◆ Slide 13: Significance of Chernoff Bound

- Much **tighter than Chebyshev**
 - Useful in:
 - Computer science
 - Network systems
 - Performance guarantees
-

◆ Slide 14: Application to Amazon Case Study

Scenario

- Millions of users accessing system simultaneously

Goal

- Estimate probability of:
 - Server overload
 - Failure conditions

Approach

- Model traffic as random variable
 - Apply tail bounds to:
 - Predict extreme loads
 - Ensure system safety margins
-

◆ Slide 15: System Reliability

Definition

System reliability is the probability that a system performs correctly under given conditions.

Connection with Tail Bounds

- Tail bounds help quantify:
 - Failure probability
 - Risk of overload
-

◆ Slide 16: Practical Insight

- Exact prediction is impossible
 - But:
 - Upper bounds give **guarantees**
 - Help in **designing robust systems**
-

◆ Slide 17: Summary

- Tail bounds are essential tools in probability
 - Three main inequalities:
 - Markov → basic bound
 - Chebyshev → variance-based
 - Chernoff → exponential bound
 - Useful in real-world systems like:
 - Online platforms
 - High-traffic events
-

◆ Slide 18: Class Participation

- Discussion and engagement on:
 - Applying tail bounds
 - Interpreting results
 - Real-world system behavior
-

✓ Final Notes for Exam Preparation

- Focus on:
 - Definitions of all three inequalities
 - Assumptions for each
 - Differences between them

- Application in system reliability
- Understand **why tighter bounds (Chernoff)** are preferred in large systems.

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LECTURE 23 SCRIBE

Predicting the Impossible: Tail Bounds and System Reliability

(Case Study: Amazon Black Friday Sale 2026 – Coding Session: Tails and Bounds)

◆ Slide 1: Lecture Introduction

Title

Predicting the Impossible: Tail Bounds and System Reliability

Case Study

Amazon Black Friday Sale 2026

Instructor Details

- Dhaval Patel, PhD
- Associate Professor, Computer Science and Engineering
- MICxN Research Lab
- SEAS, Ahmedabad University

Lecture Focus

- Understanding **tail bounds**
 - Applying probability to **real-world system reliability**
-

◆ Slide 2: Amazon Black Friday Sale 2026

Key Information

- Expected Date: **27 November 2026 (Friday)**
- Based on:
 - Historical trends
 - Expert forecasts

Context

- Large-scale event with **massive uncertainty**
 - High demand and unpredictable user traffic
-

◆ Slide 3: Connecting Computing with Uncertainty

Core Idea

- Real-world systems operate under **uncertainty**
- User behavior is **random and unpredictable**

Problem

- Cannot exactly predict:
 - Number of users

- System load
 - Failure conditions
-

◆ Slide 4: Need for Probabilistic Analysis

Why Required?

- Deterministic models fail in large-scale systems
 - Need probabilistic tools to:
 - Estimate extreme cases
 - Ensure reliability
-

◆ Slide 5: Introduction to Tail Bounds

Definition

Tail bounds are mathematical tools that provide **upper bounds** on the probability that a random variable deviates significantly from its expected value.

Purpose

- Measure risk of **extreme events**
 - Help in **system design and safety**
-

◆ Slide 6: Types of Tail Bounds

The lecture focuses on three major inequalities:

1. Markov's Inequality
 2. Chebyshev's Inequality
 3. Chernoff Bound
-

◆ Slide 7: Markov's Inequality (Definition & Formula)

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Definition

For a non-negative random variable X and $a > 0$, the probability that X exceeds a is bounded by the ratio of its expectation to a .

Assumptions

- $X \geq 0$
 - Expected value exists
-

◆ Slide 8: Derivation of Markov's Inequality (Standard)

Let $X \geq 0$

$$E[X] = \int_0^{\infty} P(X \geq t) dt$$

Since $P(X \geq a)$ is constant for $t \leq a$:

$$\begin{aligned} E[X] &\geq \int_0^a P(X \geq a) dt = a \cdot P(X \geq a) \\ \Rightarrow P(X \geq a) &\leq \frac{E[X]}{a} \end{aligned}$$

◆ Slide 9: Interpretation of Markov's Inequality

- Provides a **simple upper bound**
 - Requires only **mean**
 - Produces **loose bounds**
-

◆ Slide 10: Chebyshev's Inequality (Definition & Formula)

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Definition

For any random variable with mean μ and variance σ^2 , the probability of deviation from the mean is bounded.

Assumptions

- Mean exists
 - Variance exists
-

◆ Slide 11: Derivation of Chebyshev's Inequality (Standard)

Apply Markov's inequality to $Y = (X - \mu)^2$:

$$P((X - \mu)^2 \geq k^2 \sigma^2) \leq \frac{E[(X - \mu)^2]}{k^2 \sigma^2}$$

Since:

$$\begin{aligned} E[(X - \mu)^2] &= \sigma^2 \\ \Rightarrow P(|X - \mu| \geq k\sigma) &\leq \frac{1}{k^2} \end{aligned}$$

◆ Slide 12: Interpretation of Chebyshev's Inequality

- Measures spread around mean
- Works for **any distribution**
- More informative than Markov

◆ Slide 13: Chernoff Bound (Concept & Standard Form)

Definition (Standard)

For sum of independent random variables:

$$P(X \geq (1 + \delta)\mu) \leq e^{-\frac{\mu\delta^2}{3}} \quad (0 < \delta < 1)$$

Assumptions

- Variables are independent
- Typically Bernoulli trials

◆ Slide 14: Key Features of Chernoff Bound

- Provides **exponentially decreasing bounds**
- Much **tighter** than Chebyshev
- Suitable for **large-scale systems**

◆ Slide 15: Application to Amazon Case Study

System Modeling

- Let X = number of users accessing system

Goal

- Estimate probability of:
 - Server overload
 - System crash

Approach

- Model X as random variable
- Apply tail bounds to:
 - Predict extreme traffic
 - Set safe capacity limits

◆ Slide 16: System Reliability

Definition

System reliability is the probability that a system performs correctly under given conditions for a specified period.

Role of Tail Bounds

- Quantify:
 - Failure probability
 - Risk thresholds

◆ Slide 17: Key Insights

- Exact prediction is not possible
- Tail bounds provide:
 - Guarantees
 - Safety margins

Comparison

Inequality	Requirement	Tightness
Markov	Mean only	Loose
Chebyshev	Mean + Variance	Moderate

Inequality	Requirement	Tightness
Chernoff	Independence	Tight

◆ Slide 18: Class Participation

- Discussion on:
 - Application of inequalities
 - Real-world systems
 - Interpretation of bounds

✓ Final Exam-Oriented Summary

- Tail bounds are used to handle **uncertainty in large systems**
- Three key inequalities:
 - **Markov** → simplest, least information
 - **Chebyshev** → uses variance
 - **Chernoff** → strongest and most useful
- Application: **Amazon Black Friday Sale system reliability**

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(Case Study: Amazon Black Friday Sale 2026 – Coding Session: Tails and Bounds)

◆ Slide 1: Lecture Title

Topic

Predicting the Impossible: Tail Bounds and System Reliability

Case Study

Amazon Black Friday Sale 2026

Instructor

- Dhaval Patel, PhD
 - Associate Professor, CSE
 - MICxN Research Lab
 - SEAS, Ahmedabad University
-

◆ Slide 2: Amazon Black Friday Sale 2026

Details

- Expected Start Date: **27 November 2026 (Friday)**
- Based on:
 - Historical patterns
 - Forecasts from online sources

Observation

- Event involves **large-scale uncertainty**
 - Heavy traffic and unpredictable user behavior
-

◆ Slide 3: Uncertainty in Computing Systems

Concept

- Real-world systems operate under uncertainty
- User requests and traffic are random

Challenge

- Exact prediction of system load is not possible
-

◆ Slide 4: Motivation

Need

- To handle unpredictable system behavior
- To estimate extreme conditions such as overload

Approach

- Use probabilistic tools for bounding outcomes
-

◆ Slide 5: Tail Bounds

Definition

Tail bounds provide an **upper bound on the probability** that a random variable deviates significantly from its expected value.

Purpose

- Quantify risk of extreme events
 - Assist in system reliability analysis
-

◆ Slide 6: Types of Tail Bounds

The lecture introduces:

1. Markov's Inequality
 2. Chebyshev's Inequality
 3. Chernoff Bound
-

◆ Slide 7: Markov's Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Definition

For a non-negative random variable X , the probability that it exceeds a value a is bounded by the ratio of its expectation to a .

Assumptions

- $X \geq 0$
 - Expected value exists
-

◆ Slide 8: Derivation of Markov's Inequality (Standard Form)

For $X \geq 0$:

$$E[X] = \int_0^{\infty} P(X \geq t) dt$$

For $t \leq a$, $P(X \geq t) \geq P(X \geq a)$:

$$\begin{aligned} E[X] &\geq \int_0^a P(X \geq a) dt = a \cdot P(X \geq a) \\ \Rightarrow P(X \geq a) &\leq \frac{E[X]}{a} \end{aligned}$$

◆ Slide 9: Interpretation of Markov's Inequality

- Requires only mean
 - Provides loose bound
 - Useful when limited information is available
-

◆ Slide 10: Chebyshev's Inequality

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Definition

For any random variable, the probability of deviation from its mean is bounded using variance.

Assumptions

- Mean μ exists
 - Variance σ^2 exists
-

◆ Slide 11: Derivation of Chebyshev's Inequality

Apply Markov's inequality to $(X - \mu)^2$:

$$P((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{E[(X - \mu)^2]}{k^2\sigma^2}$$

Since:

$$\begin{aligned} E[(X - \mu)^2] &= \sigma^2 \\ \Rightarrow P(|X - \mu| \geq k\sigma) &\leq \frac{1}{k^2} \end{aligned}$$

◆ Slide 12: Interpretation of Chebyshev's Inequality

- Works for any distribution
 - Uses variance
 - Gives tighter bound than Markov
-

◆ Slide 13: Chernoff Bound

Standard Definition

For sum of independent random variables:

$$P(X \geq (1 + \delta)\mu) \leq e^{-\frac{\mu\delta^2}{3}} \quad (0 < \delta < 1)$$

Assumptions

- Random variables are independent
 - Often applied to Bernoulli trials
-

◆ Slide 14: Properties of Chernoff Bound

- Provides exponentially decreasing bounds
 - Much tighter than other inequalities
 - Suitable for large-scale systems
-

◆ Slide 15: Application to Amazon Case Study

Model

- Let X represent system load (number of users)

Objective

- Estimate probability of:
 - System overload
 - Failure

Method

- Apply tail bounds to predict extreme cases
-

◆ Slide 16: System Reliability

Definition

System reliability is the probability that a system performs correctly under specified conditions.

Use of Tail Bounds

- Estimate failure probabilities
- Design systems with safety margins

◆ Slide 17: Key Observations

- Exact prediction is not feasible
- Tail bounds provide **upper guarantees**

Comparison

Method	Requirement	Nature
Markov	Mean only	Loose
Chebyshev	Mean + Variance	Moderate
Chernoff	Independence	Tight

◆ Slide 18: Class Participation

- Discussion on:
 - Tail bounds application
 - System reliability
 - Interpretation of probabilistic guarantees

✓ Final Exam Notes

- Learn:
 - Definitions
 - Assumptions
 - Derivations of Markov & Chebyshev
 - Concept of Chernoff bound
- Focus on **application in system reliability (Amazon case study)**

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- ✓ This scribe is **complete, structured, and strictly based on your PDF**
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